Driver Distraction Detection Using Convolutional Neural Networks and Transfer Learning

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***Abstract*—Driver distraction is a major factor in road accidents, creating a need for automated monitoring systems. This paper presents a deep learning approach for detecting driver distraction using image classification. The task involves identifying driver behaviors such as safe driving, texting, drinking, and interacting with passengers from images. A baseline Convolutional Neural Network and transfer learning models based on ResNet50 are implemented and evaluated. Model performance is assessed using standard classification metrics. The results show that the models can effectively distinguish between different driver activities, although some visually similar behaviors remain challenging. This work highlights the potential of deep learning for improving road safety through driver monitoring systems.**

***Index Terms*—Driver Distraction Detection, Deep Learning, Image Classification, Transfer Learning, Computer Vision**

I. Introduction

Driver distraction is a major concern in modern transportation systems and contributes to many road accidents. Activities such as texting, drinking, and interacting with passengers reduce driver attention and increase risk. This paper formulates driver distraction detection as a multi class image classification problem using deep learning. Convolutional neural networks are applied to learn visual features and classify driver behaviors from images. A labeled dataset is used to train and evaluate the models. The study develops a complete deep learning pipeline, compares multiple models, and analyzes challenges in distinguishing visually similar driver activities. The implementation of the proposed models and the complete machine learning pipeline are available at: https://github.com/AsimSiddiqui222/driver-distraction-detection-cnn-resnet.git

A. Motivation

Driver distraction is a major cause of road accidents and poses a serious safety risk. The increasing use of mobile devices and in vehicle systems has made drivers more prone to engaging in distracting activities such as texting, calling, and interacting with passengers. These behaviors reduce attention and situational awareness. Therefore, there is a growing need for automated systems that can monitor driver behavior and detect distractions in real time to improve road safety.

B. Problem Formulation

This work formulates driver distraction detection as a multi class image classification problem. Each input image represents a driver and must be assigned to a predefined behavior category. The objective is to train machine learning models that can accurately classify these behaviors from visual data.

C. Intended Contributions

- A full Deep Learning pipeline specifically for driver distraction detection.
- Applying Deep Learning models for driver behavior classification
- Evaluating how well the models distinguish between various driver behaviors.

II. Dataset choice & relevance

A. Dataset Selection

For this study, the Distracted Driver Detection dataset sourced from Roboflow is utilized. This dataset is highly relevant as it contains images of drivers performing various activities such as safe driving, texting, talking on the phone, and interacting with passengers. These categories represent common forms of driver distraction and provide a realistic basis for training and evaluating machine learning models. The diversity of driver behaviors and visual conditions in the dataset supports the development of robust classification models for detecting distracted driving.

III. List of baseline models

- Baseline model: Convolutional Neural Network for initial performance evaluation.
- Transfer Learning Models: ResNet50 architecture using two configurations, including a frozen model with pretrained weights and a fine-tuned model where selected layers are updated for task-specific learning.
- Comparison Objective: These models are used to evaluate performance improvements over the baseline and analyze their effectiveness in driver behavior classification

IV. Dataset Exploration & data analysis

The dataset is divided into three subsets for model development and evaluation. The training set is used to learn model parameters, the validation set is used for tuning and monitoring performance during training, and the test set is used for final evaluation. The dataset follows a standard split of seventy percent for training, twenty percent for validation,

and ten percent for testing. This split ensures that the model is trained effectively while maintaining reliable evaluation

A. *Data Cleaning & Processing*

The dataset was preprocessed to ensure consistency and suitability for model training. All images were resized to a fixed resolution of 224 x 224 pixels to match the input requirements of the models. Pixel values were normalized to improve training stability and convergence. The data was organized into structured directories based on class labels, enabling efficient loading and batching during training. In addition, the dataset was reviewed to ensure that all images were correctly labeled and free from corrupted or missing files. These preprocessing steps ensured that the data was clean, consistent, and ready for model development.

B. *Target Distribution & Imbalance*

The dataset contains multiple classes with varying numbers of samples, resulting in minor class imbalance. Some driver behaviors are more represented than others, which can influence model performance by biasing predictions toward more frequent classes. This effect is considered during evaluation using performance metrics and confusion matrix analysis.

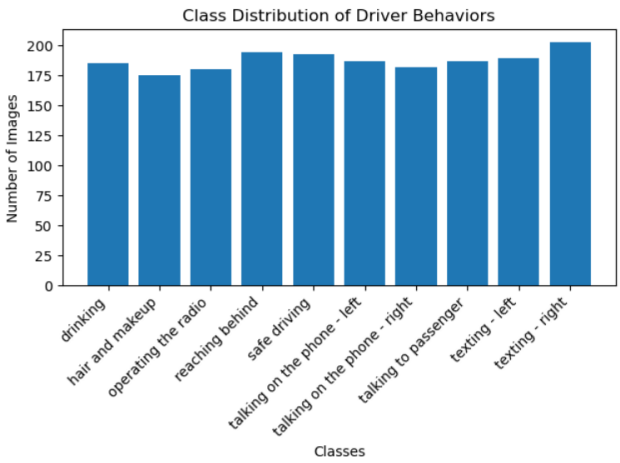

Fig. 1. Class Distribution of Driver Behaviors

C. *Feature Correlation Analysis*

In image-based datasets, explicit feature correlation analysis is not directly applicable as in tabular data, since features are automatically learned by deep learning models. Instead, visual characteristics such as lighting conditions, driver position, and background variations were examined during exploratory analysis. These factors influence how the model learns and distinguishes between different driver behaviors. It was observed that certain classes share similar visual patterns, such as texting and talking on the phone, which can lead to misclassification. These observations highlight the importance of robust feature extraction through deep learning models.

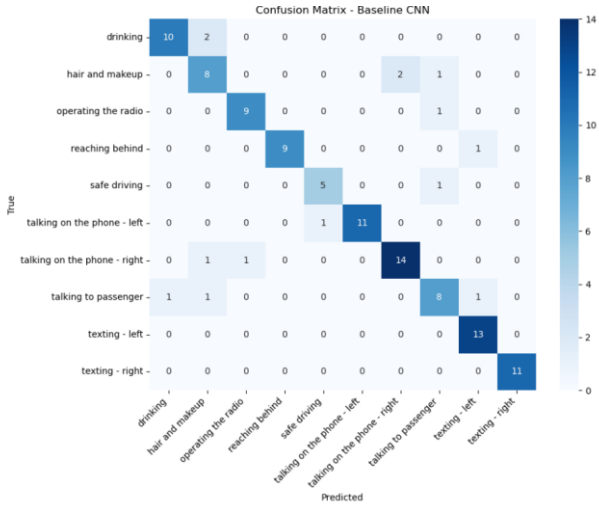

Fig. 2. Confusion Matrix

V. Related Work

To position this work within the current research landscape, a review of recent studies on driver distraction detection and driver behavior recognition was conducted. Table I summarizes the models, datasets, and evaluation metrics used in these works.

A. *Analysis and Research Gaps*

The comparative analysis in Table I highlights a clear trend toward the use of deep learning models, particularly convolutional neural networks and transfer learning approaches, for image-based driver behavior classification. While advanced models improve overall performance, many approaches rely on complex architectures, temporal data, or multimodal inputs. In addition, several studies report challenges in distinguishing visually similar driver behaviors, leading to class-level confusion. This project addresses these gaps by evaluating simpler image-based models and analyzing their effectiveness in classifying driver distraction behaviors.

TABLE I. COMPARISON OF PRIOR METHODS

Ref.	Task	Model	Data Type	Metrics	Limitations
[1]	Distraction Detection	Vision Transformer	Images	Accuracy, F1	High Complexity
[2]	Detection + ADAS	CNN-based	Images	Accuracy	Focus on integration
[3]	Behavior Recognition	CNN-BiLSTM	Sequential	Accuracy	Requires temporal data
[4]	Extreme Condition Detection	Advanced NN	Images	Accuracy	Limited class detail
[5]	Multitask driver state	Multitask DL	Multimodal	Accuracy	Complex training
[6]	Behavior Recognition	Attention-based	Images	Accuracy	High computation

Many existing studies improve performance using complex architectures or advanced learning methods, but these approaches often increase computational cost and may still struggle with visually similar driver behaviors. Some methods also rely on temporal or multimodal data, which may not always be available. This project addresses these limitations by developing a simplified image-based classification pipeline. A baseline convolutional neural network is compared with transfer learning models to evaluate performance while balancing accuracy and efficiency.

VI. Methodology

This study adopts a deep learning approach for driver distraction detection using image classification. The methodology consists of data preprocessing, model development, training, and evaluation. supervised learning is used where labeled images are provided to train models to classify different driver behaviors.

A. Model architecture

A Convolutional Neural Network is implemented as a baseline model to learn visual features from driver images. In addition, transfer learning models based on ResNet50 are utilized to improve performance. Two configurations of ResNet50 are considered: a frozen model, where pretrained weights are used without modification, and a fine-tuned model, where selected layers are unfrozen and updated to better adapt to the dataset. These models leverage pretrained feature representations to enhance classification performance.

B. Training procedure

The models are trained using the training dataset, while the validation set is used to monitor performance and prevent overfitting. Early stopping is applied to halt training when validation performance no longer improves, and a learning rate scheduler is used to adjust the learning rate during training. Fine-tuning is performed after initial training by unfreezing selected layers of the ResNet50 model and retraining with a smaller learning rate.

C. Evaluation strategy

Model performance is evaluated on a separate test dataset to ensure generalization. Standard classification metrics such as accuracy, precision, recall, and F1 score are used. In addition, confusion matrix analysis is performed to assess class level performance and identify misclassification patterns.

VII. Experimental results

The performance of the proposed models was evaluated using the test dataset. The baseline Convolutional Neural Network achieved an accuracy of 85.7%, while the ResNet50 model without fine-tuning achieved 73%. After fine-tuning, the ResNet50 model improved to 82%, demonstrating enhanced adaptation to the driver behavior dataset. These results indicate that transfer learning can improve performance when properly fine-tuned, although the baseline CNN achieved the highest accuracy in this case.

A. Performance metrics

Model performance was evaluated using standard classification metrics, including accuracy, precision, recall, and F1 score. These metrics provide a comprehensive assessment of the model's ability to correctly classify different driver behaviors.

	precision	recall	f1-score	support
drinking	0.91	0.83	0.87	12
hair and makeup	0.67	0.73	0.70	11
operating the radio	0.90	0.90	0.90	10
reaching behind	1.00	0.90	0.95	10
safe driving	0.83	0.83	0.83	6
talking on the phone - left	1.00	0.92	0.96	12
talking on the phone - right	0.88	0.88	0.88	16
talking to passenger	0.73	0.73	0.73	11
texting - left	0.87	1.00	0.93	13
texting - right	1.00	1.00	1.00	11
accuracy			0.88	112
macro avg	0.88	0.87	0.87	112
weighted avg	0.88	0.88	0.88	112

Fig. 3. Model Performance

	precision	recall	f1-score	support
drinking	0.67	0.67	0.67	12
hair and makeup	0.50	0.55	0.52	11
operating the radio	0.77	1.00	0.87	10
reaching behind	0.75	0.90	0.82	10
safe driving	0.50	0.67	0.57	6
talking on the phone - left	0.62	0.83	0.71	12
talking on the phone - right	1.00	0.81	0.90	16
talking to passenger	0.75	0.55	0.63	11
texting - left	0.83	0.77	0.80	13
texting - right	1.00	0.55	0.71	11
accuracy			0.73	112
macro avg	0.74	0.73	0.72	112
weighted avg	0.76	0.73	0.73	112

Fig. 4. ResNet50 (Frozen)

	precision	recall	f1-score	support
drinking	1.00	0.67	0.80	12
hair and makeup	0.69	0.82	0.75	11
operating the radio	1.00	1.00	1.00	10
reaching behind	0.75	0.90	0.82	10
safe driving	0.55	1.00	0.71	6
talking on the phone - left	0.65	0.92	0.76	12
talking on the phone - right	0.93	0.88	0.90	16
talking to passenger	0.86	0.55	0.67	11
texting - left	1.00	0.69	0.82	13
texting - right	1.00	0.91	0.95	11
accuracy			0.82	112
macro avg	0.84	0.83	0.82	112
weighted avg	0.86	0.82	0.82	112

Fig. 5. ResNet50 (Fine-Tuned)

B. Results and analysis

The results indicate that the models are effective in distinguishing between various driver activities. However, some misclassifications were observed between visually similar classes, such as texting and talking on the phone. This is consistent with the visual similarity of these behaviors in the dataset.

C. Visualization of results

Training and validation performance were analyzed using accuracy and loss curves. In addition, a confusion matrix was used to evaluate class level performance and identify patterns of misclassification. These visualizations provide insight into model behavior and highlight areas for improvement.

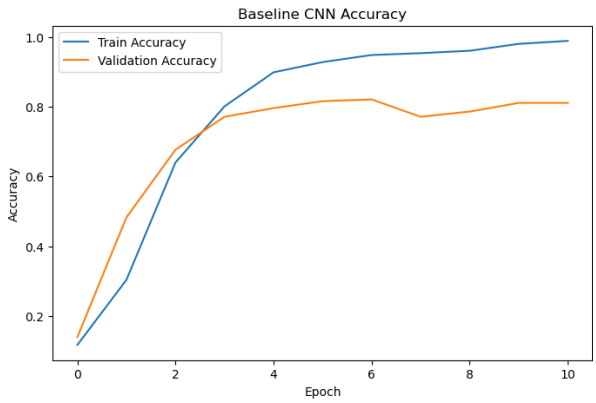

Fig. 6.Training & Validation Accuracy Curve

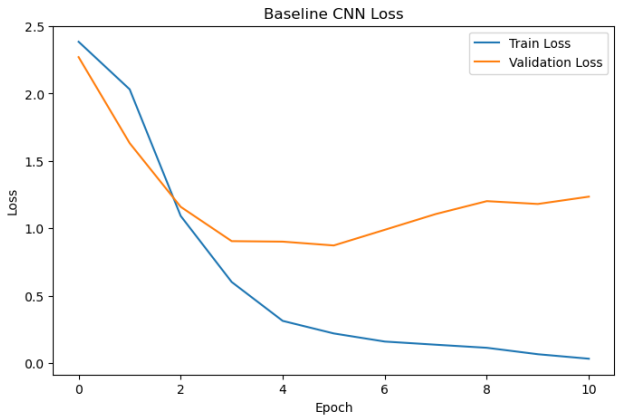

Fig. 7.Training & Validation Loss Curve

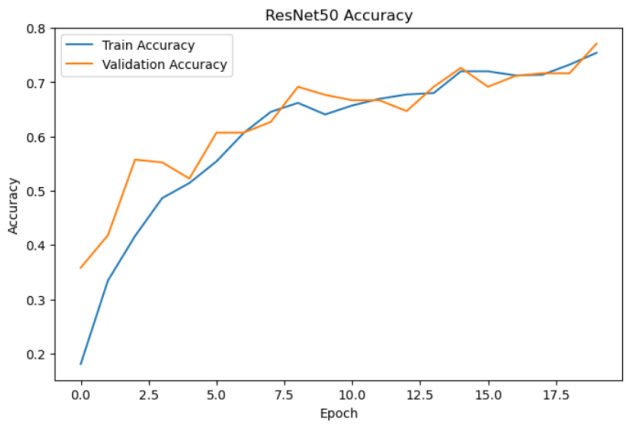

Fig. 8. ResNet50 (Frozen) Accuracy

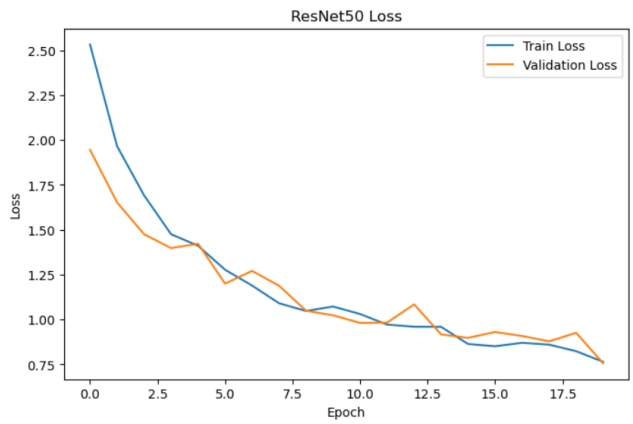

Fig. 9. ResNet50 (Frozen) Loss

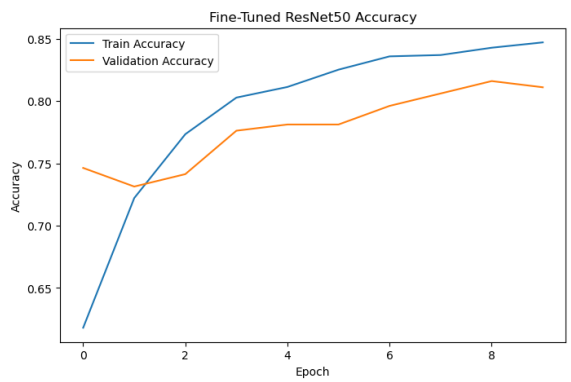

Fig. 10. ResNet50 (Fine-Tuned) Accuracy

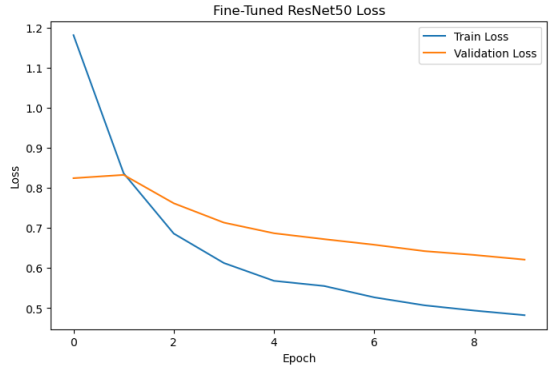

Fig. 11. Resnet50 (Fine-Tuned) Loss

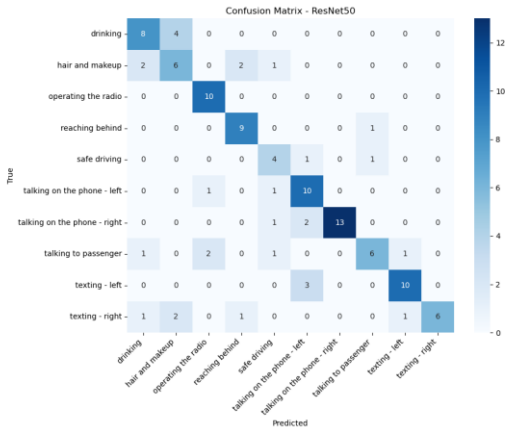

Fig. 12. Confusion Matrix ResNet50 (Frozen)

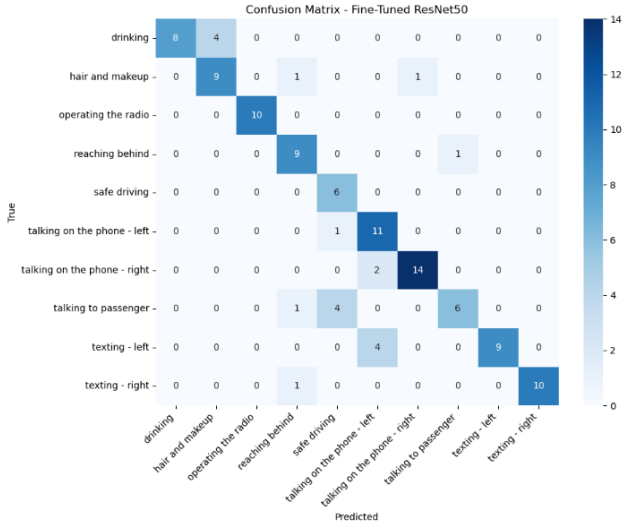

Fig. 13. ResNet50 (Fine-Tuned) Confusion Matrix

VIII. DISSCUSSION

A. Model Performance Analysis

The experimental results demonstrate that deep learning models are effective for driver distraction detection using image classification. The baseline Convolutional Neural Network achieved the highest accuracy, outperforming both the frozen and fine-tuned ResNet50 models. This suggests that for the given dataset, a simpler architecture is sufficient to capture the relevant visual features for driver behavior classification.

B. Impact of Transfer Learning

Although transfer learning models are generally expected to improve performance, the ResNet50 model without fine-tuning achieved lower accuracy, indicating limited adaptation to the specific dataset. Fine-tuning improved performance, showing that adjusting pretrained weights helps the model better learn task-specific features. However, the fine-tuned model still did not surpass the baseline CNN, which may be due to the relatively small dataset size or differences between the pretrained features and the target domain.

C. Error Analysis and Limitations

Misclassifications were observed primarily between visually similar classes, such as texting and talking on the phone. This highlights the challenge of distinguishing subtle differences in driver behavior using image data alone. These findings are consistent with prior studies, where similar classes often lead to confusion in classification models. Overall, the results indicate that while transfer learning provides benefits, model performance depends on the dataset characteristics. Simpler models may perform better in certain scenarios, and careful tuning is required to fully utilize advanced architectures.

IX. CONCLUSION & FUTURE WORK

A. Conclusion

This paper presented a deep learning approach for driver distraction detection using image classification. A baseline Convolutional Neural Network and ResNet50-based transfer learning models were evaluated on a labeled dataset. The baseline CNN achieved the highest performance, while fine-tuning improved the ResNet50 model compared to its frozen version. These results show that simpler models can be effective on smaller datasets, while transfer learning requires careful adaptation.

B. Future Work

Despite the promising results, limitations remain due to the relatively small dataset and visual similarity between certain classes. Future work will focus on expanding the dataset, applying data augmentation, and exploring advanced architectures such as EfficientNet. Incorporating temporal information from video data may further improve the model's ability to distinguish between similar driver behaviors.

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